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Axial Green's function method for multi-dimensional elliptic boundary value problems

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SUMMARY

Axial Green's function method is proposed to solve multi-dimensional second-order elliptic partial differential equations with the geometrical complexity of solution domains. Instead of directly dealing with 2D or 3D Green's functions, the proposed method systematically decomposes a multi-dimensional elliptic differential operator into 1D ones. Consequently, the method only requires 1D Green's function associated with each coordinate axis. Theoretical formulation and simple axial discretization result in the system of equations of which the solutions represent the numerical solution of the original multi-dimensional elliptic problem, which is otherwise intractable to obtain by the traditional multi-dimensional Green's function technique. The method employs analytical form of 1D Green's function and integral schemes so that it can potentially be more robust than both the finite difference method and the boundary element method. In addition, the simple discretization procedure, which uses only axially orthogonal straight lines and boundary data, demonstrates its convenience over the finite element method of comparable accuracy. Numerical examples verify these advantages by demonstrating the second-order convergence in various discretized models of complex geometries. Copyright © 2008 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Green's function method has long been applied to boundary value problems for a variety of physical phenomena [1–9]. Given an inhomogeneous partial differential equation with boundary

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conditions, its unique solution is represented by the convolution integral of the Green's function that has the same singularity of the corresponding fundamental solution and satisfies the associated homogeneous boundary conditions [10]. For each particular boundary value problem, the availability of the fundamental solution is thus a key to the existence of the Green's function and its applicability. The Laplace equation, the heat equation with constant coefficient, and the Helmholtz equation in a homogeneous medium [11–13] are among examples that the fundamental solutions are available for the Green's function technique.

In general, it is however intractable to find a fundamental solution and the Green's function for an arbitrary multi-dimensional elliptic partial differential equation with the complex geometry of the solution domain [14–16]. Only a few exceptions, in conjunction with simple domains such as a circle or a sphere, can be explored by this analytical method. In order to circumvent such difficulties, some attempts have been made to obtain numerical Green's functions [17–21], but high cost for computing is still inevitable. In contrast, if we focus on 1D boundary value problems, then the Green's function method can be applied to a much broader class of problems because the closed forms of 1D fundamental solution and the corresponding Green's function are obtainable with more ease [22, 23].

In this paper, we propose a novel numerical method that can solve general multi-dimensional second-order elliptic boundary value problems with variable coefficients, by employing 1D Green's function only. The multi-dimensional differential operator is first decomposed axially, and then a system of 1D problems, which is equivalent to the original higher-dimensional boundary value problem, is derived by introducing new variables associated with the respective axial dimensions. Each of the 1D problems has the representation formula for the Green's function solution defined on axially straight lines. After eliminating the original solution variable from the system, we are able to obtain integral equations for those new axially associated variables, at any desired point in the interior of the solution domain. Through simple axial discretization, these integral equations constitute a system of equations. The original solution variable is finally determined from the axial variables that are the solutions of the system of equations represented by the 1D axial Green's functions.

The solution domain is discretized by a group of axial lines (i.e. straight lines parallel with the respective axes) and their intersecting points, as well as the boundary points at which the axial lines meet the boundary. The spacing between the lines can be arbitrary, and the method is thus able to facilitate any complex geometry without difficulty, which is the apparent advantage over the finite difference method [24]. Moreover, the derived system of equations consists of integral equations so that the method can be viewed as an integral scheme. This notion is an additional advantage comparable with the finite element method [25]. In general, the method achieves the $O(h^2)$ convergence for the solution and, for its derivative, the convergence higher than $O(h)$, regardless of the regularity in discretization. Together, the integral scheme and the geometrical flexibility make the method robust in terms of convergence and adaptability, which will be demonstrated through several numerical examples later in this paper.

It is necessary here to address the difference between the proposed method and the boundary element method (BEM), both of which might seem to have similar features. However, in BEM, the fundamental solution has to be essentially obtainable in an exact form [26–32]. As mentioned above, it is generally very difficult to find the fundamental solution or the Green's function for any 2D or 3D second-order elliptic problem, except for a few differential operators such as the Laplacian. This causes severe restrictions on the range of problems to which BEM can practically be applied. However, the current method only requires 1D Green's functions for the axially

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decomposed 1D operators in conjunction with straightforward domain discretization. Therefore, the proposed method does not suffer from such a limitation of applicability.

The paper is presented as follows. We begin with a brief review of the Green's function approach for 1D second-order elliptic problems in the following section. In Section 3, the proposed formulations for 2D and 3D problems are developed as the major theoretical foundations of the method. Discretization for computer implementation and the resultant system of equations are described in Section 4. Numerical examples are presented in Sections 5–7, respectively, for 1D, 2D, and 3D examples. Finally, concluding remarks in Section 8 will complete the paper.

2. GREEN'S FUNCTION THEORY FOR 1D SECOND-ORDER ELLIPTIC PROBLEMS

To begin with, here we review 1D Green's function theory for 1D elliptic problems because it is the fundamental building block of the current method to solve multi-dimensional problems. Let us assume an open interval, $I=(a,b)$ in $\mathbb{R}^1$, where $\kappa(t)>0$ is continuous and $1/\kappa(t)$ is square integrable in I. We consider the following elliptic problem in $\mathbb{R}^1$:

$$(\kappa u_t)_t = \phi, \quad t \in I \tag{1}$$

$$u(a) = u^a, \quad u(b) = u^b \tag{2}$$

where the subscript t denotes the derivative with respect to t. If τ is an arbitrary point in I, the Green's function for this elliptic problem is defined as the solution of the following differential equation:

$$(\kappa(t)\, G_t(t,\tau))_t = \delta(t-\tau) \quad \text{for all } t \in I \tag{3}$$

$$G(a,\tau) = G(b,\tau) = 0 \tag{4}$$

where $\delta(t-\tau)$ is the Dirac delta function as a function of t with the point singularity at τ. The exact solution of the differential equation (3), satisfying the homogeneous boundary conditions (4), is obtained as

$$G(t,\tau) = \begin{cases} -\dfrac{K(a,t)K(\tau,b)}{K(a,b)}, & t<\tau \\ -\dfrac{K(a,\tau)K(t,b)}{K(a,b)}, & t \geqslant \tau \end{cases} \tag{5}$$

where the function K is defined as an integral of the reciprocal of κ:

$$K(u,v) = \int_u^v \frac{1}{\kappa(s)}\, \mathrm{d}s \tag{6}$$

We summarize the properties of the Green's function $G(t,\tau)$ in Equation (5) as follows:

(i) *Dirac delta property and transmission condition*:

$$[k(t)G_t(t,\tau)]_{t=\tau} = 1, \quad [k(t)G_t(t,\tau)]_{t\neq\tau} = 0 \tag{7}$$

where the bracket implies $[\bullet]_{t=\alpha} = \lim_{t\to\alpha^+} \bullet - \lim_{t\to\alpha^-} \bullet$.

(ii) *Continuity*: $G(t,\tau)$ is continuous in I with respect to the variable t.

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(iii) *Discontinuity of the derivative*:

$$G_t(t,\tau)=\begin{cases}-\dfrac{1}{\kappa(t)}\dfrac{K(\tau,b)}{K(a,b)}, & t<\tau\\ \dfrac{1}{\kappa(t)}\dfrac{K(a,\tau)}{K(a,b)}, & t\geqslant\tau\end{cases} \tag{8}$$

(iv) *Pointwise homogeneity*:

$$(\kappa(t)G_t(t,\tau))_t=0, \quad t\neq\tau \tag{9}$$

Using the Green's function $G(t,\tau)$, the solution of the 1D elliptic problem (3) is represented as

$$u(\tau)=\int_a^b G(t,\tau)\phi(t)\,\mathrm{d}t+\kappa(b)\,G_t(b,\tau)u(b)-\kappa(a)\,G_t(a,\tau)u(a), \quad \tau\in I \tag{10}$$

Applying the non-homogeneous boundary conditions (2), the solution $u(\tau)$ can equivalently be rewritten as

$$u(\tau)=\int_a^b G(t,\tau)\phi(t)\,\mathrm{d}t+u_b\frac{K(a,\tau)}{K(a,b)}+u_a\frac{K(\tau,b)}{K(a,b)}, \quad \tau\in I \tag{11}$$

and accordingly its derivative is given as

$$u'(\tau)=\frac{1}{\kappa(\tau)}\left\{\int_a^b\left[\frac{K(a,t)}{K(a,b)}-H(t-\tau)\right]\phi(t)\,\mathrm{d}t+\frac{u_b-u_a}{K(a,b)}\right\} \tag{12}$$

where H denotes the Heaviside function.

The key operator in the representation formula (11) is the convolution integral associated with the Green's function, which is defined on the space of square integrable functions in I, $L^2(I)$, such that

$$\Theta(\phi)(\tau)=\int_I G(t,\tau)\phi(t)\,\mathrm{d}t, \quad \tau\in I \tag{13}$$

Then, by investigating the general properties of this operator, we were able to show that the solution in Equation (11) is the unique weak solution of the 1D elliptic problem, Equations (1) and (2), under the condition that $k>0$ is continuous and $1/k$ is square integrable.

For practical computations of the solution in Equation (11), we need to approximate $\phi(t)$ by means of basis functions such as continuous 1D linear finite element shape functions. Let $\phi_h(t)$ be one of such approximations and $u_h(t)$ be the corresponding numerical solution. Then, the error between $u(t)$ and $u_h(t)$ can be estimated by the Schwarz inequality:

$$\begin{aligned}\|u-u_h\|_\infty &\leqslant \sqrt{b-a}\max_{\tau\in I}\frac{K(a,\tau)K(\tau,b)}{K(a,b)}\,\|\phi-\phi_h\|_2\\ &\leqslant \frac{\sqrt{b-a}}{4}K(a,b)\,\|\phi-\phi_h\|_2\leqslant\frac{b-a}{4}\left\|\frac{1}{\kappa}\right\|_2\|\phi-\phi_h\|_2\end{aligned} \tag{14}$$

where $\|\bullet\|_2$ is the L^2 norm and $\|\bullet\|_\infty=\sup_{t\in I}|\bullet|$. If we use the exact $\phi(x)$, instead of $\phi_h(x)$, the numerical integration would then be the only source of errors.

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3. FORMULATION FOR MULTI-DIMENSIONAL SECOND-ORDER ELLIPTIC PROBLEMS

Let us now consider second-order elliptic partial differential equations in an open domain Ω and the boundary condition on $\partial\Omega$, in higher-dimensional space $\mathbb{R}^n$ $(n=2,3)$, as

$$\nabla\cdot(\kappa\nabla u)=f \quad \text{in } \Omega \tag{15}$$

$$u=u^{\partial\Omega} \quad \text{on } \partial\Omega \tag{16}$$

where $\kappa(\mathbf{x})\,(>0)$ is continuous, and $1/\kappa$ and $f(\mathbf{x})$ are axially square integrable functions in Ω. The axial square integrability of a function in $\Omega\subset\mathbb{R}^n$ means that the function is square integrable on every axis-parallel line segment in Ω. The following two subsections detail our method to solve the elliptic partial differential equation (15) satisfying the Dirichlet boundary condition (16) in 2D and 3D, respectively, using the 1D Green's function described in Section 2.

3.1. 2D second-order elliptic problems

Let $\mathbf{x}=(x,y)\in\mathbb{R}^2$. By introducing a new variable $\phi(\mathbf{x})$, the problem stated in Equations (15) and (16) can be split as

$$(\kappa u_x)_x=\phi(x,y) \quad \text{in } \Omega \tag{17}$$

$$(\kappa u_y)_y=f(x,y)-\phi(x,y) \quad \text{in } \Omega \tag{18}$$

$$u=u^{\partial\Omega} \quad \text{on } \partial\Omega \tag{19}$$

where we must solve for two unknowns, ϕ and u. However, we will show that the 1D Green's function enables us to reduce the number of unknowns down to only one, i.e. ϕ. This is the main idea of the proposed formulation.

Let us first consider an arbitrary solution domain as shown in Figure 1. Assume $X_{\bar{y}}$ and $Y_{\bar{x}}$ are axial line segments, parallel to x-axis and y-axis, respectively. They are connected at and pass axially through the point $(\bar{x},\bar{y})\in\Omega$, and their end points are located on $\partial\Omega$:

$$X_{\bar{y}}=\{(x,\bar{y})\in\Omega \mid x_-<x<x_+,\ (x_\pm,\bar{y})\in\partial\Omega\} \tag{20}$$

$$Y_{\bar{x}}=\{(\bar{x},y)\in\Omega \mid y_-<y<y_+,\ (\bar{x},y_\pm)\in\partial\Omega\} \tag{21}$$

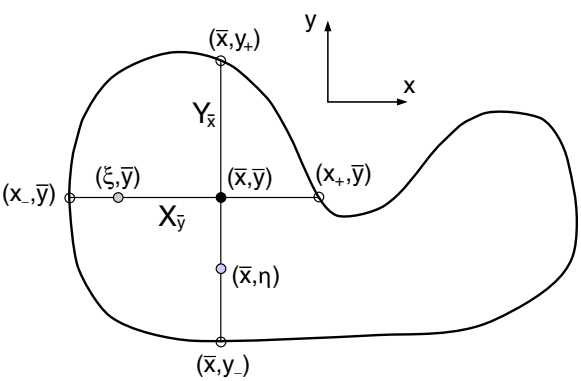

Figure 1. Illustration of axial line segments parallel with each axis, passing through a point $(\bar{x},\bar{y})\in\Omega$.

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The boundary points of axial line segments are denoted by

$$\partial X_{\bar{y}} = \{(x_-, \bar{y}), (x_+, \bar{y})\}, \quad \partial Y_{\bar{x}} = \{(\bar{x}, y_-), (\bar{x}, y_+)\} \tag{22}$$

Although $x_\pm$ and $y_\pm$ depend on $\bar{y}$ and $\bar{x}$, respectively, we often omit the dependence unless confusing.

On axial line segments, $X_{\bar{y}}$ and $Y_{\bar{x}}$, for a fixed $(\bar{x}, \bar{y})$, the 1D Green's functions $G(x, \xi; X_{\bar{y}})$ and $G(y, \eta; Y_{\bar{x}})$ can, respectively, be defined as the solutions of the following ordinary differential equations:

$$(\kappa(x, \bar{y})\, G_x(x, \xi; X_{\bar{y}}))_x = \delta(x - \xi), \quad G(\partial X_{\bar{y}}, \xi; X_{\bar{y}}) = 0 \tag{23}$$

$$(\kappa(\bar{x}, y)\, G_y(y, \eta; Y_{\bar{x}}))_y = \delta(y - \eta), \quad G(\partial Y_{\bar{x}}, \eta; Y_{\bar{x}}) = 0 \tag{24}$$

where ξ denotes the x-coordinate of an arbitrary point on $X_{\bar{y}}$, η the y-coordinate of an arbitrary point on $Y_{\bar{x}}$. $\delta(x-\xi)$ and $\delta(y-\eta)$ are 1D Dirac delta functions on axial line segments, $X_{\bar{y}}$ and $Y_{\bar{x}}$, with respective point singularities, $(\xi, \bar{y}) \in X_{\bar{y}}$, and $(\bar{x}, \eta) \in Y_{\bar{x}}$. Equivalently, they satisfy the following equations:

$$(\kappa(x, \bar{y}) G_x(x, \xi; X_{\bar{y}}))_x = 0 \quad \text{when } x \neq \xi \tag{25}$$

$$\kappa(\xi^+, \bar{y}) G_x(\xi^+, \xi; X_{\bar{y}}) - \kappa(\xi^-, \bar{y}) G_x(\xi^-, \xi; X_{\bar{y}}) = 1 \tag{26}$$

$$G(\xi^+, \xi; X_{\bar{y}}) - G(\xi^-, \xi; X_{\bar{y}}) = 0 \tag{27}$$

$$G(\partial X_{\bar{y}}, \xi; X_{\bar{y}}) = 0 \tag{28}$$

and

$$(\kappa(\bar{x}, y) G_y(y, \eta; Y_{\bar{x}}))_y = 0 \quad \text{when } y \neq \eta \tag{29}$$

$$\kappa(\bar{x}, \eta^+) G_y(\eta^+, \eta; Y_{\bar{x}}) - \kappa(\bar{x}, \eta^-) G_y(\eta^-, \eta; Y_{\bar{x}}) = 1 \tag{30}$$

$$G(\eta^+, \eta; Y_{\bar{x}}) - G(\eta^-, \eta; Y_{\bar{x}}) = 0 \tag{31}$$

$$G(\partial Y_{\bar{x}}, \eta; Y_{\bar{x}}) = 0 \tag{32}$$

By solving these ordinary differential equations, we obtain the 1D Green's functions that are basically the same as those introduced in Section 2:

$$G(x, \xi; X_{\bar{y}}) = \begin{cases} -\dfrac{K(x_-, x; X_{\bar{y}}) K(\xi, x_+; X_{\bar{y}})}{K(x_-, x_+; X_{\bar{y}})}, & x < \xi \\ -\dfrac{K(x_-, \xi; X_{\bar{y}}) K(x, x_+; X_{\bar{y}})}{K(x_-, x_+; X_{\bar{y}})}, & x \geqslant \xi \end{cases} \tag{33}$$

$$G(y, \eta; Y_{\bar{x}}) = \begin{cases} -\dfrac{K(y_-, y; Y_{\bar{x}}) K(\eta, y_+; Y_{\bar{x}})}{K(y_-, y_+; Y_{\bar{x}})}, & y < \eta \\ -\dfrac{K(y_-, \eta; Y_{\bar{x}}) K(y, y_+; Y_{\bar{x}})}{K(y_-, y_+; Y_{\bar{x}})}, & y \geqslant \eta \end{cases} \tag{34}$$

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where we call $G(x,\xi;X_{\bar{y}})$ and $G(y,\eta;Y_{\bar{x}})$ the axial Green's functions on $X_{\bar{y}}$ and $Y_{\bar{x}}$, respectively. In Equations (33) and (34), $K(\alpha,\beta;X_{\bar{y}})$ and $K(\alpha,\beta;Y_{\bar{x}})$ are 1D integrals defined on $X_{\bar{y}}$ and $Y_{\bar{x}}$ as

$$K(\alpha,\beta;X_{\bar{y}}) \equiv \int_{\alpha}^{\beta} \frac{1}{\kappa(\tau,\bar{y})}\,\mathrm{d}\tau \tag{35}$$

$$K(\alpha,\beta;Y_{\bar{x}}) \equiv \int_{\alpha}^{\beta} \frac{1}{\kappa(\bar{x},\tau)}\,\mathrm{d}\tau \tag{36}$$

From the axial Green's functions in Equations (33) and (34), we have the following representations for the solution $u(x,y)$ of Equations (17) , (18), and (19):

$$\begin{aligned} u(\xi,\bar{y}) = &\int_{x_-}^{x_+} G(\tau,\xi;X_{\bar{y}})\phi(\tau,\bar{y})\,\mathrm{d}\tau \\ &+\kappa(x_+,\bar{y})u(x_+,\bar{y})G_x(x_+,\xi;X_{\bar{y}}) - \kappa(x_-,\bar{y})u(x_-,\bar{y})G_x(x_-,\xi;X_{\bar{y}}) \end{aligned} \tag{37}$$

$$\begin{aligned} u(\bar{x},\eta) = &\int_{y_-}^{y_+} G(\tau,\eta;Y_{\bar{x}})(f(\bar{x},\tau)-\phi(\bar{x},\tau))\,\mathrm{d}\tau \\ &+\kappa(\bar{x},y_+)u(\bar{x},y_+)G_y(y_+,\eta;Y_{\bar{x}}) - \kappa(\bar{x},y_-)u(\bar{x},y_-)G_y(y_-,\eta;Y_{\bar{x}}) \end{aligned} \tag{38}$$

If using the derivatives of Equations (33) and (34) as

$$G_x(x,\xi;X_{\bar{y}}) = \begin{cases} -\dfrac{1}{\kappa(x,\bar{y})}\dfrac{K(\xi,x_+;X_{\bar{y}})}{K(x_-,x_+;X_{\bar{y}})}, & x<\xi \\ \dfrac{1}{\kappa(x,\bar{y})}\dfrac{K(x_-,\xi;X_{\bar{y}})}{K(x_-,x_+;X_{\bar{y}})}, & x>\xi \end{cases} \tag{39}$$

$$G_y(y,\eta;Y_{\bar{x}}) = \begin{cases} -\dfrac{1}{\kappa(\bar{x},y)}\dfrac{K(\eta,y_+;Y_{\bar{x}})}{K(y_-,y_+;Y_{\bar{x}})}, & y<\eta \\ \dfrac{1}{\kappa(\bar{x},y)}\dfrac{K(y_-,\eta;Y_{\bar{x}})}{K(y_-,y_+;Y_{\bar{x}})}, & y>\eta \end{cases} \tag{40}$$

and setting $\xi=\bar{x}$ in Equation (37) and $\eta=\bar{y}$ in Equation (38), then the integral equation for the unknown $\phi(x,y)$ can be obtained as

$$\begin{aligned} &\int_{x_-}^{x_+} G(\tau,\bar{x};X_{\bar{y}})\phi(\tau,\bar{y})\,\mathrm{d}\tau \\ &\quad+\frac{K(x_-,\bar{x};X_{\bar{y}})}{K(x_-,x_+;X_{\bar{y}})}u(x_+,\bar{y}) + \frac{K(\bar{x},x_+;X_{\bar{y}})}{K(x_-,x_+;X_{\bar{y}})}u(x_-,\bar{y}) \\ &\quad=\int_{y_-}^{y_+} G(\tau,\bar{y};Y_{\bar{x}})[f(\bar{x},\tau)-\phi(\bar{x},\tau)]\,\mathrm{d}\tau \\ &\qquad+\frac{K(y_-,\bar{y};Y_{\bar{x}})}{K(y_-,y_+;Y_{\bar{x}})}u(\bar{x},y_+) + \frac{K(\bar{y},y_+;Y_{\bar{x}})}{K(y_-,y_+;Y_{\bar{x}})}u(\bar{x},y_-) \end{aligned} \tag{41}$$

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for all $(\bar{x}, \bar{y}) \in \Omega$. If we rearrange terms with respect to ϕ, we finally have

$$\begin{aligned}
&\int_{x_-}^{x_+} G(\tau, \bar{x}; X_{\bar{y}})\phi(\tau, \bar{y})\,\mathrm{d}\tau + \int_{y_-}^{y_+} G(\tau, \bar{y}; Y_{\bar{x}})\phi(\bar{x}, \tau)\,\mathrm{d}\tau \\
&= \int_{y_-}^{y_+} G(\tau, \bar{y}; Y_{\bar{x}})\, f(\bar{x}, \tau)\,\mathrm{d}\tau + \frac{K(y_-, \bar{y}; Y_{\bar{x}})}{K(y_-, y_+; Y_{\bar{x}})} u(\bar{x}, y_+) + \frac{K(\bar{y}, y_+; Y_{\bar{x}})}{K(y_-, y_+; Y_{\bar{x}})} u(\bar{x}, y_-) \\
&\quad - \frac{K(x_-, \bar{x}; X_{\bar{y}})}{K(x_-, x_+; X_{\bar{y}})} u(x_+, \bar{y}) - \frac{K(\bar{x}, x_+; X_{\bar{y}})}{K(x_-, x_+; X_{\bar{y}})} u(x_-, \bar{y})
\end{aligned} \tag{42}$$

for all $(\bar{x}, \bar{y}) \in \Omega$. We will call this equation *the cross-integral equation for* ϕ. Once ϕ is obtained, we can compute the solution of the original partial differential equation, u, by the representation formula with ϕ as follows:

$$\begin{aligned}
u(\bar{x}, \bar{y}) = &\int_{x_-}^{x_+} G(\tau, \bar{x}; X_{\bar{y}})\phi(\tau, \bar{y})\,\mathrm{d}\tau \\
&+\kappa(x_+, \bar{y}) u(x_+, \bar{y})\, G_x(x_+, \bar{x}; X_{\bar{y}}) \\
&-\kappa(x_-, \bar{y}) u(x_-, \bar{y})\, G_x(x_-, \bar{x}; X_{\bar{y}}), \quad \bar{x} \in X_{\bar{y}}
\end{aligned} \tag{43}$$

$$\begin{aligned}
u(\bar{x}, \bar{y}) = &\int_{y_-}^{y_+} G(\tau, \bar{y}; Y_{\bar{x}})(f(\bar{x}, \tau) - \phi(\bar{x}, \tau))\,\mathrm{d}\tau \\
&+\kappa(\bar{x}, y_+) u(\bar{x}, y_+) G_y(y_+, \bar{y}; Y_{\bar{x}}) \\
&-\kappa(\bar{x}, y_-) u(\bar{x}, y_-) G_y(y_-, \bar{y}; Y_{\bar{x}}), \quad \bar{y} \in Y_{\bar{x}}
\end{aligned} \tag{44}$$

Consequently, we represent the solution $u(x, y)$ in terms of the axial Green's function and the source function, both of which are associated with any line segment parallel to each axis.

3.2. 3D second-order elliptic problems

We now develop the axial decomposition of the 3D operator into three 1D parts, similar to Equations (17) and (18). For 3D problems, we need to introduce two new variables ϕ_1 and ϕ_2 to separate the elliptic operator such that

$$(\kappa u_x)_x = \phi_1(x, y, z) \quad \text{in } \Omega \tag{45}$$

$$(\kappa u_y)_y = \phi_2(x, y, z) \quad \text{in } \Omega \tag{46}$$

$$(\kappa u_z)_z = f(x, y, z) - \phi_1(x, y, z) - \phi_2(x, y, z) \quad \text{in } \Omega \tag{47}$$

$$u = u^{\partial\Omega} \quad \text{on } \partial\Omega \tag{48}$$

$X_{\bar{y}\bar{z}}$, $Y_{\bar{x}\bar{z}}$, and $Z_{\bar{x}\bar{y}}$ are the axial line segments associated with a point $(\bar{x}, \bar{y}, \bar{z})$. The axial representations of the solution are thus expressed by means of the three axial Green's functions, $G(x, \xi; X_{\bar{y}\bar{z}})$,

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$G(y,\eta;Y_{\bar{x}\bar{z}})$, and $G(z,\zeta;Z_{\bar{x}\bar{y}})$, respectively, on $X_{\bar{y}\bar{z}}$, $Y_{\bar{x}\bar{z}}$, and $Z_{\bar{x}\bar{y}}$, as follows:

$$\begin{aligned} u(\xi,\bar{y},\bar{z}) = &\int_{x_-}^{x_+} G(\tau,\xi;X_{\bar{y}\bar{z}})\phi_1(\tau,\bar{y},\bar{z})\,\mathrm{d}\tau \\ &+\kappa(x_+,\bar{y},\bar{z})u(x_+,\bar{y},\bar{z})G_x(x_+,\xi;X_{\bar{y}\bar{z}}) \\ &-\kappa(x_-,\bar{y},\bar{z})u(x_-,\bar{y},\bar{z})G_x(x_-,\xi;X_{\bar{y}\bar{z}}) \end{aligned} \tag{49}$$

$$\begin{aligned} u(\bar{x},\eta,\bar{z}) = &\int_{y_-}^{y_+} G(\tau,\eta;Y_{\bar{x}\bar{z}})\phi_2(\bar{x},\tau,\bar{z})\,\mathrm{d}\tau \\ &+\kappa(\bar{x},y_+,\bar{z})u(\bar{x},y_+,\bar{z})G_y(y_+,\eta;Y_{\bar{x}\bar{z}}) \\ &-\kappa(\bar{x},y_-,\bar{z})u(\bar{x},y_-,\bar{z})G_y(y_-,\eta;Y_{\bar{x}\bar{z}}) \end{aligned} \tag{50}$$

$$\begin{aligned} u(\bar{x},\bar{y},\zeta) = &\int_{z_-}^{z_+} G(\tau,\zeta;Z_{\bar{x}\bar{y}})(f(\bar{x},\bar{y},\tau)-\phi_1(\bar{x},\bar{y},\tau)-\phi_2(\bar{x},\bar{y},\tau))\,\mathrm{d}\tau \\ &+\kappa(\bar{x},\bar{y},z_+)u(\bar{x},\bar{y},z_+)G_z(z_+,\zeta;Z_{\bar{x}\bar{y}}) \\ &-\kappa(\bar{x},\bar{y},z_-)u(\bar{x},\bar{y},z_-)G_z(z_-,\zeta;Z_{\bar{x}\bar{y}}) \end{aligned} \tag{51}$$

where $x_\pm$, $y_\pm$, and $z_\pm$, respectively, denote x-, y-, and z-coordinates of the end points of the three axial line segments, $X_{\bar{y}\bar{z}}$, $Y_{\bar{x}\bar{z}}$, and $Z_{\bar{x}\bar{y}}$, associated with the point $(\bar{x},\bar{y},\bar{z})$.

At the point $(\xi,\eta,\zeta)=(\bar{x},\bar{y},\bar{z})$, all of the u's on the left-hand sides in the above Equations, (49), (50), and (51), become identical. Therefore, by equating them, we can finally construct two integral equations. For example, from Equation (51) = Equation (49) and Equation (51) = Equation (50), we have

$$\begin{aligned} &\int_{x_-}^{x_+} G(\tau,\bar{x};X_{\bar{y}\bar{z}})\phi_1(\tau,\bar{y},\bar{z})\,\mathrm{d}\tau+\int_{z_-}^{z_+} G(\tau,\bar{z};Z_{\bar{x}\bar{y}})(\phi_1(\bar{x},\bar{y},\tau)+\phi_2(\bar{x},\bar{y},\tau))\,\mathrm{d}\tau \\ &=\int_{z_-}^{z_+} G(\tau,\bar{z};Z_{\bar{x}\bar{y}})f(\bar{x},\bar{y},\tau)\,\mathrm{d}\tau \\ &\quad+\frac{K(z_-,\bar{z};Z_{\bar{x}\bar{y}})}{K(z_-,z_+;Z_{\bar{x}\bar{y}})}u(\bar{x},\bar{y},z_+)+\frac{K(\bar{z},z_+;Z_{\bar{x}\bar{y}})}{K(z_-,z_+;Z_{\bar{x}\bar{y}})}u(\bar{x},\bar{y},z_-) \\ &\quad-\frac{K(x_-,\bar{x};X_{\bar{y}\bar{z}})}{K(x_-,x_+;X_{\bar{y}\bar{z}})}u(x_+,\bar{y},\bar{z})-\frac{K(\bar{x},x_+;X_{\bar{y}\bar{z}})}{K(x_-,x_+;X_{\bar{y}\bar{z}})}u(x_-,\bar{y},\bar{z}) \end{aligned} \tag{52}$$

$$\begin{aligned} &\int_{y_-}^{y_+} G(\tau,\bar{y};Y_{\bar{x}\bar{z}})\phi_2(\bar{x},\tau,\bar{z})\,\mathrm{d}\tau+\int_{z_-}^{z_+} G(\tau,\bar{z};Z_{\bar{x}\bar{y}})(\phi_1(\bar{x},\bar{y},\tau)+\phi_2(\bar{x},\bar{y},\tau))\,\mathrm{d}\tau \\ &=\int_{z_-}^{z_+} G(\tau,\bar{z};Z_{\bar{x}\bar{y}})f(\bar{x},\bar{y},\tau)\,\mathrm{d}\tau \\ &\quad+\frac{K(z_-,\bar{z};Z_{\bar{x}\bar{y}})}{K(z_-,z_+;Z_{\bar{x}\bar{y}})}u(\bar{x},\bar{y},z_+)+\frac{K(\bar{z},z_+;Z_{\bar{x}\bar{y}})}{K(z_-,z_+;Z_{\bar{x}\bar{y}})}u(\bar{x},\bar{y},z_-) \\ &\quad-\frac{K(y_-,\bar{y};Y_{\bar{x}\bar{z}})}{K(y_-,y_+;Y_{\bar{x}\bar{z}})}u(\bar{x},y_+,\bar{z})-\frac{K(\bar{y},y_+;Y_{\bar{x}\bar{z}})}{K(y_-,y_+;Y_{\bar{x}\bar{z}})}u(\bar{x},y_-,\bar{z}) \end{aligned} \tag{53}$$

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for any $(\bar{x}, \bar{y}, \bar{z}) \in \Omega$. The representation formula of 3D problem are finally expressed as

$$\begin{aligned} u(\bar{x}, \bar{y}, \bar{z}) = & \int_{x_-}^{x_+} G(\tau, \bar{x}; X_{\bar{y}\bar{z}}) \phi_1(\tau, \bar{y}, \bar{z}) \, \mathrm{d}\tau \\ & + \kappa(x_+, \bar{y}, \bar{z}) u(x_+, \bar{y}, \bar{z}) G_x(x_+, \bar{x}; X_{\bar{y}\bar{z}}) \\ & - \kappa(x_-, \bar{y}, \bar{z}) u(x_-, \bar{y}, \bar{z}) G_x(x_-, \bar{x}; X_{\bar{y}\bar{z}}), \quad x \in X_{\bar{y}\bar{z}} \end{aligned} \tag{54}$$

$$\begin{aligned} u(\bar{x}, \bar{y}, \bar{z}) = & \int_{y_-}^{y_+} G(\tau, \bar{y}; Y_{\bar{x}\bar{z}}) \phi_2(\bar{x}, \tau, \bar{z}) \, \mathrm{d}\tau \\ & + \kappa(\bar{x}, y_+, \bar{z}) u(\bar{x}, y_+, \bar{z}) G_y(y_+, \bar{y}; Y_{\bar{x}\bar{z}}) \\ & - \kappa(\bar{x}, y_-, \bar{z}) u(\bar{x}, y_-, \bar{z}) G_y(y_-, \bar{y}; Y_{\bar{x}\bar{z}}), \quad y \in Y_{\bar{x}\bar{z}} \end{aligned} \tag{55}$$

$$\begin{aligned} u(\bar{x}, \bar{y}, \bar{z}) = & \int_{z_-}^{z_+} G(\tau, \bar{z}; Z_{\bar{x}\bar{y}}) (f(\bar{x}, \bar{y}, \tau) - \phi_1(\bar{x}, \bar{y}, \tau) - \phi_2(\bar{x}, \bar{y}, \tau)) \, \mathrm{d}\tau \\ & + \kappa(\bar{x}, \bar{y}, z_+) u(\bar{x}, \bar{y}, z_+) G_z(z_+, \bar{z}; Z_{\bar{x}\bar{y}}) \\ & - \kappa(\bar{x}, \bar{y}, z_-) u(\bar{x}, \bar{y}, z_-) G_z(z_-, \bar{z}; Z_{\bar{x}\bar{y}}), \quad x \in Z_{\bar{x}\bar{y}} \end{aligned} \tag{56}$$

for any $(\bar{x}, \bar{y}, \bar{z}) \in \Omega$. As a consequence, the solution $u(x, y, z)$ is represented by the axial Green's functions and the source function that are associated with any line segment parallel to each axis.

4. AXIAL DISCRETIZATION OF INTEGRAL EQUATIONS

In this section, we describe the discretization of the resultant 1D integral equation (42) on each axial line. For convenience, here we deal with the 2D domain only as shown in Figure 2. However, this description can straightforwardly be extended to 3D axial discretization of which a numerical example will be presented later in this paper. Assume Ω to be an open-bounded domain in $\mathbb{R}^2$ and $\mathbf{x} = (x, y)$ a point in $\mathbb{R}^2$. The function ϕ is the unique unknown in the integral equation (42). Once ϕ is obtained, the solution u is computed by the representation formula defined in Equations (43) and (44). Let X_j and Y_i be straight line segments in Ω parallel with x-axis and y-axis, respectively. They commonly pass through a point $(\bar{x}_i, \bar{y}_j) \in \Omega$ as illustrated in Figure 2, i.e.

$$X_j \equiv \{(x, \bar{y}_j) | (x, \bar{y}_j) \in \Omega\}, \quad Y_i \equiv \{(\bar{x}_i, y) | (\bar{x}_i, y) \in \Omega\} \tag{57}$$

We call the node $(\bar{x}_i, \bar{y}_j) \in \Omega$ *an* (i, j) *base point* in Ω.

Remark 1
There happen to exist multiple axial lines, X_j's (or Y_i's), for common j (or i) if the domain is not convex as can be noticed in Figure 2. They should be treated as individual axial lines. However, in this section, we simply use one notation, X_j (or Y_i), in order to avoid any complex notations and clearly present the key idea.

First, in order to generate the base points $(\bar{x}_i, \bar{y}_j) \in \Omega$ on which the unknown function ϕ will be discretized, let us assume a larger rectangular region $[x_L, x_R] \times [y_B, y_T]$ that entirely covers Ω.

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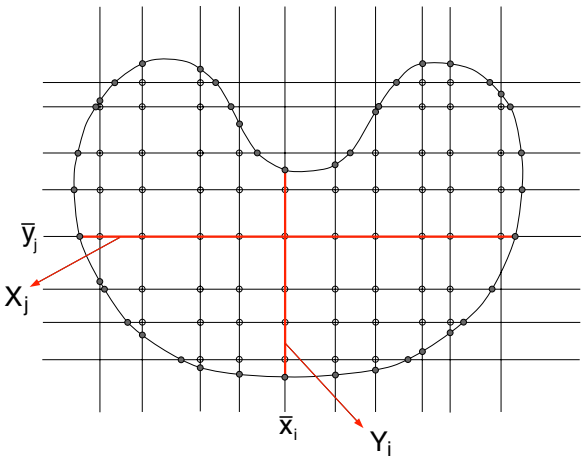

Figure 2. Discretized 2D axial line segments X_j and Y_i of which the base point is $(\bar{x}_i, \bar{y}_j)$ in $\Omega \in \mathbb{R}^2$.

We divide the intervals, $[x_L, x_R]$ and $[y_B, y_T]$, into partitions, $x_L = x_0 < x_1 < \cdots < x_n = x_R$ and $y_B = y_0 < y_1 < \cdots < y_m = y_T$, respectively. Among them, we identify all of the points (x_{i^*}, y_{j^*}) that are located in the interior of Ω. Then, these x_{i^*}'s and y_{j^*}'s are renumbered to be

$$\bar{x}_1 < \bar{x}_2 < \cdots < \bar{x}_{\bar{n}}, \quad \bar{y}_1 < \bar{y}_2 < \cdots < \bar{y}_{\bar{m}} \tag{58}$$

for appropriate $\bar{n}$ and $\bar{m}$, which are not greater than n and m, respectively.

Let Λ^{O} be the set of all base points in Ω, i.e.

$$\Lambda^{\mathrm{O}} \equiv \{(\bar{x}_i, \bar{y}_j) \in \Omega | 1 \leqslant i \leqslant \bar{n},\ 1 \leqslant j \leqslant \bar{m}\} \tag{59}$$

For the effective description of the axial discretization, we define the collinear base points on X_j and Y_i, respectively, denoted by

$$\Lambda_j \equiv X_j \cap \Lambda^{\mathrm{O}}, \quad j = 1, \ldots, \bar{m} \tag{60}$$

$$\Lambda_i \equiv Y_i \cap \Lambda^{\mathrm{O}}, \quad i = 1, \ldots, \bar{n} \tag{61}$$

and the boundary points for axial line segments, $\partial\Lambda_j$ and $\partial\Lambda_i$, as

$$\partial\Lambda_j \equiv \{(x_-, \bar{y}_j), (x_+, \bar{y}_j)\}, \quad j = 1, \ldots, \bar{m} \tag{62}$$

$$\partial\Lambda_i \equiv \{(\bar{x}_i, y_-), (\bar{x}_i, y_+)\}, \quad i = 1, \ldots, \bar{n} \tag{63}$$

From these definitions, we have

$$\Lambda^{\mathrm{O}} = \bigcup_{j=1}^{\bar{m}} \Lambda_j = \bigcup_{i=1}^{\bar{n}} \Lambda_i \tag{64}$$

$$\Lambda^{\mathrm{B}} = \left(\bigcup_{j=1}^{\bar{m}} \partial\Lambda_j \right) \cup \left(\bigcup_{i=1}^{\bar{n}} \partial\Lambda_i \right) \tag{65}$$

where Λ^{B} denotes the set of boundary nodes.

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Based on this discretization procedure, the unknown ϕ can be approximated on each axial line segment in terms of 1D piecewise linear interpolation. That is, let us assume that the 1D partitions of X_j and Y_i with respect to Λ_j and Λ_i are, respectively, given as

$$X_j = \{(x_-, \bar{y}_j) < (\bar{x}_{i^*}, \bar{y}_j) < (\bar{x}_{i^*+1}, \bar{y}_j) < \cdots < (\bar{x}_{n^*}, \bar{y}_j) < (x_+, \bar{y}_j)\} \tag{66}$$

$$Y_i = \{(\bar{x}_i, y_-) < (\bar{x}_i, \bar{y}_{j^*}) < (\bar{x}_i, \bar{y}_{j^*+1}) < \cdots < (\bar{x}_i, \bar{y}_{m^*}) < (\bar{x}_i, y_+)\} \tag{67}$$

where i^*, n^* and j^*, m^* depend on specific $\bar{y}_j$ and $\bar{x}_i$, respectively. The approximation of ϕ is then defined as follows: for $i^*-1 \leqslant k \leqslant n^*$ and $j^*-1 \leqslant s \leqslant m^*$,

$$\phi^h(\xi, \bar{y}_j) = \phi(\bar{x}_k, \bar{y}_j)\frac{\xi - \bar{x}_{k+1}}{\bar{x}_k - \bar{x}_{k+1}} + \phi(\bar{x}_{k+1}, \bar{y}_j)\frac{\bar{x}_k - \xi}{\bar{x}_k - \bar{x}_{k+1}}, \quad \xi \in [\bar{x}_k, \bar{x}_{k+1}] \tag{68}$$

$$\phi^h(\bar{x}_i, \eta) = \phi(\bar{x}_i, \bar{y}_s)\frac{\eta - \bar{y}_{s+1}}{\bar{y}_s - \bar{y}_{s+1}} + \phi(\bar{x}_i, \bar{y}_{s+1})\frac{\bar{y}_s - \eta}{\bar{y}_s - \bar{y}_{s+1}}, \quad \eta \in [\bar{y}_s, \bar{y}_{s+1}] \tag{69}$$

for all $i = 1, \ldots, \bar{n}$ and $j = 1, \ldots, \bar{m}$. Here, we imply $\bar{x}_{i^*-1} = x_-$, $\bar{x}_{n^*+1} = x_+$, $\bar{y}_{j^*-1} = y_-$, and $\bar{y}_{m^*+1} = y_+$.

By replacing ϕ with this piecewise linear approximation ϕ^h and collecting terms in terms of nodal solution of ϕ, we can convert the original integral equation of Equation (42) into the system of equations as follows:

$$\begin{aligned}
&\phi(x_-, \bar{y}_j)\underbrace{\int_{x_-}^{\bar{x}_{i^*}} G(\tau, \bar{x}_i; X_j)\frac{\tau - \bar{x}_{i^*}}{x_- - \bar{x}_{i^*}}\,\mathrm{d}\tau}_{I} + \phi(x_+, \bar{y}_j)\underbrace{\int_{\bar{x}_{n^*}}^{x_+} G(\tau, \bar{x}_i; X_j)\frac{\bar{x}_{n^*} - \tau}{\bar{x}_{n^*} - x_+}\,\mathrm{d}\tau}_{II} \\
&\quad + \sum_{k=i^*}^{n^*(\bar{y}_j)} \phi(\bar{x}_k, \bar{y}_j)\left[\int_{\bar{x}_k}^{\bar{x}_{k+1}} G(\tau, \bar{x}_i; X_j)\frac{\tau - \bar{x}_{k+1}}{\bar{x}_k - \bar{x}_{k+1}}\,\mathrm{d}\tau + \int_{\bar{x}_{k-1}}^{\bar{x}_k} G(\tau, \bar{x}_i; X_j)\frac{\bar{x}_{k-1} - \tau}{\bar{x}_{k-1} - \bar{x}_k}\,\mathrm{d}\tau\right] \\
&\quad + \phi(\bar{x}_i, y_-)\underbrace{\int_{y_-}^{\bar{y}_{j^*}} G(\tau, \bar{y}_j; Y_i)\frac{\tau - \bar{y}_{j^*}}{y_- - \bar{y}_{j^*}}\,\mathrm{d}\tau}_{III} + \phi(\bar{x}_i, y_+)\underbrace{\int_{\bar{y}_{m^*}}^{y_+} G(\tau, \bar{y}_j; Y_i)\frac{\bar{y}_{m^*} - \tau}{\bar{y}_{m^*} - y_+}\,\mathrm{d}\tau}_{IV} \\
&\quad + \sum_{s=j^*}^{m^*(\bar{x}_i)} \phi(\bar{x}_i, \bar{y}_s)\left[\int_{\bar{y}_s}^{\bar{y}_{s+1}} G(\tau, \bar{y}_j; Y_i)\frac{\tau - \bar{y}_{s+1}}{\bar{y}_s - \bar{x}_{s+1}}\,\mathrm{d}\tau + \int_{\bar{y}_{s-1}}^{\bar{y}_s} G(\tau, \bar{y}_j; Y_i)\frac{\bar{y}_{s-1} - \tau}{\bar{y}_{s-1} - \bar{y}_s}\,\mathrm{d}\tau\right] \\
&\quad = \int_{y_-}^{y_+} G(\tau, \bar{y}_j; Y_i) f(\bar{x}_i, \tau)\,\mathrm{d}\tau \\
&\qquad + \frac{K(y_-, \bar{y}_j; Y_i)}{K(y_-, y_+; Y_i)} u(\bar{x}_i, y_+) + \frac{K(\bar{y}_j, y_+; Y_i)}{K(y_-, y_+; Y_i)} u(\bar{x}_i, y_-) \\
&\qquad - \frac{K(x_-, \bar{x}_i; X_j)}{K(x_-, x_+; X_j)} u(x_+, \bar{y}_j) - \frac{K(\bar{x}_i, x_+; X_j)}{K(x_-, x_+; X_j)} u(x_-, \bar{y}_j)
\end{aligned} \tag{70}$$

for all $(\bar{x}_i, \bar{y}_j) \in \Lambda^{\mathrm{O}}$. For the 1D integrals that appeared in Equation (70), we employ, in the numerical examples of the following sections, the trapezoidal rule or the two-point Gaussian quadrature as a numerical integration scheme of the convergence higher than second order.

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In the system of Equations (70), the integrals, I, II, III, and IV are the coefficients of the boundary nodal values of ϕ. If we use the Trapezoidal rule, these integrals vanish, because the Green's functions have zero values at the boundary and the linear functions in I, II, III, and IV vanish at the first inward base point from the boundary. Thus, the system of Equation (70) is completed. On the other hand, if using two-point Gaussian quadrature rule for the integrals, those terms do not vanish. In this case, we can also drop out the terms, based on the following second-order error estimate:

$$
\begin{aligned}
|I| &\leqslant h\left|\int_0^1 t\,G(\bar{x}_{i^*}+t(x_- - \bar{x}_{i^*}),\bar{x}_i;X_j)\,\mathrm{d}t\right| \\
&= h\frac{K(\bar{x}_i,x_+;X_j)}{K(x_-,x_+;X_j)}\int_0^1 t\,K(x_-,\bar{x}_{i^*}+t(x_- - \bar{x}_{i^*});X_j)\,\mathrm{d}t \\
&\leqslant \left(\frac{1}{\sqrt{20}}\frac{K(\bar{x}_i,x_+;X_j)}{K(x_-,x_+;X_j)}\left\|\frac{1}{\kappa}\right\|_2\right)h^2 \leqslant \left(\frac{1}{\sqrt{20}}\left\|\frac{1}{\kappa}\right\|_2\right)h^2
\end{aligned}
\tag{71}
$$

where $\|\bullet\|_2$ is the L^2 norm and $h=|\bar{x}_{i^*}-x_-|$. Similarly, we can show that the other integrals, II, III, and IV, have errors of $O(h^2)$ where h is the distance between the boundary point and the first inward base point analogous to the case of I.

Remark 2
If we define matrices G^{X_j} and G^{Y_i} associated with (70),

$$
G_{ik}^{X_j} = \int_{\bar{x}_k}^{\bar{x}_{k+1}} G(\tau,\bar{x}_i;X_j)\frac{\tau-\bar{x}_{k+1}}{\bar{x}_k-\bar{x}_{k+1}}\,\mathrm{d}\tau + \int_{\bar{x}_{k-1}}^{\bar{x}_k} G(\tau,\bar{x}_i;X_j)\frac{\bar{x}_{k-1}-\tau}{\bar{x}_{k-1}-\bar{x}_k}\,\mathrm{d}\tau \tag{72}
$$

$$
G_{js}^{Y_i} = \int_{\bar{y}_s}^{\bar{y}_{s+1}} G(\tau,\bar{y}_j;Y_i)\frac{\tau-\bar{y}_{s+1}}{\bar{y}_s-\bar{x}_{s+1}}\,\mathrm{d}\tau + \int_{\bar{y}_{s-1}}^{\bar{y}_s} G(\tau,\bar{y}_j;Y_i)\frac{\bar{y}_{s-1}-\tau}{\bar{y}_{s-1}-\bar{y}_s}\,\mathrm{d}\tau \tag{73}
$$

then we have the following matrix equation

$$
\sum_k G_{ik}^{X_j}\phi_{kj} + \sum_s G_{js}^{Y_i}\phi_{is} = F_{ij} + B_{ij}, \quad (\bar{x}_i,\bar{y}_j)\in\Lambda^{\mathrm{O}} \tag{74}
$$

where F_{ij} and B_{ij} correspond to two kinds of terms on the right-hand side of (70). Calculating the matrices G^{X_j} and G^{Y_i} requires numerous numerical integrations. This procedure ensures the robustness in convergence, although it is a time-consuming part.

5. 1D NUMERICAL EXAMPLE

As a preliminary numerical example, we first study in this section the solution convergence of the 1D problem. In the following two sections (Sections 6 and 7), we will solve several 2D and 3D examples. Let us consider the 1D elliptic partial differential equation such as

$$
(\kappa(x)u_x(x))_x = \phi(x), \quad x\in I=(-0.5,0.5) \tag{75}
$$

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where

$$\kappa(x)=2+\frac{2}{\pi}\tan^{-1}\left(\frac{-2x-x^2}{\varepsilon}\right) \quad \text{and} \quad \varepsilon=\frac{1}{20} \tag{76}$$

The boundary conditions, $u(-0.5)$ and $u(0.5)$, are given from the analytical solution as

$$u(x)=(1+x^2)\kappa(x) \tag{77}$$

The variable coefficient $\kappa(x)$ has a steep change near the origin and so does $u(x)$ according to Equation (77), which is our intent to choose this 1D example.

If ϕ is exact and the integrations in Equation (11) are analytically performed, the numerical solution will perfectly match with the exact solution (and its derivatives also). Nevertheless, we will use the approximate ϕ in Equation (11) so that we can investigate the integral operator Θ and the solution behavior for this 1D example. It is because our main purpose is to examine the proposed approximate method for multi-dimensional problems of which the exact forms are not available.

First, let us assume the exact solution $u(x)$ is provided in formula (11). Then, we discretize the integral operator Θ by approximating ϕ in terms of piecewise linear polynomial on the axial line as in Equation (68). Consequently, the resultant system of equations for the unknown ϕ is obtained. Two numerical integration schemes (i.e. the trapezoidal rule and the two-point Gaussian quadrature rule) are employed for the comparison of convergence. For the 1D example, we initially distribute the nodes uniformly in $[-1, 1]$, which covers the problem domain $I=(-0.5, 0.5)$. Then, by extracting out the nodes inside I and attaching the boundary points of I to them, the union results in the set of nodal points on the axial line $[-0.5, 0.5]$.

By solving the discretized systems for several cases with different nodal distances (h), we investigate the convergence of the nodal solutions of ϕ. The derivative of the solution, $\mathrm{d}u/\mathrm{d}x$, is also calculated, based on the derivative formula (12). The results are shown in Figure 3. The convergence results for ϕ and $\mathrm{d}u/\mathrm{d}x$ are given in Figure 4, demonstrating the second order of convergence for ϕ from both numerical integration schemes. In the case of the two-point Gaussian quadrature, the convergence of $\mathrm{d}u/\mathrm{d}x$ is almost of third order, which is apparently higher than that of trapezoidal rule. The procedure to obtain ϕ for a given u is in fact the differentiation, $(\kappa u_x)_x$, and we attain the second-order convergence for ϕ by using the piecewise linear approximation only. Therefore, this example demonstrates the better performance of the proposed axial Green's function method over the finite difference method.

6. 2D NUMERICAL EXAMPLES

For 2D examples, we consider the elliptic partial differential equation as

$$\nabla\cdot(\kappa(x,y)\nabla u(x,y))=f(x,y), \quad (x,y)\in\Omega \tag{78}$$

where

$$\kappa(x,y)=2+\frac{2}{\pi}\tan^{-1}\left(\frac{-2x-y^2+y-x^2}{\varepsilon}\right) \quad \text{and} \quad \varepsilon=\frac{1}{10} \tag{79}$$

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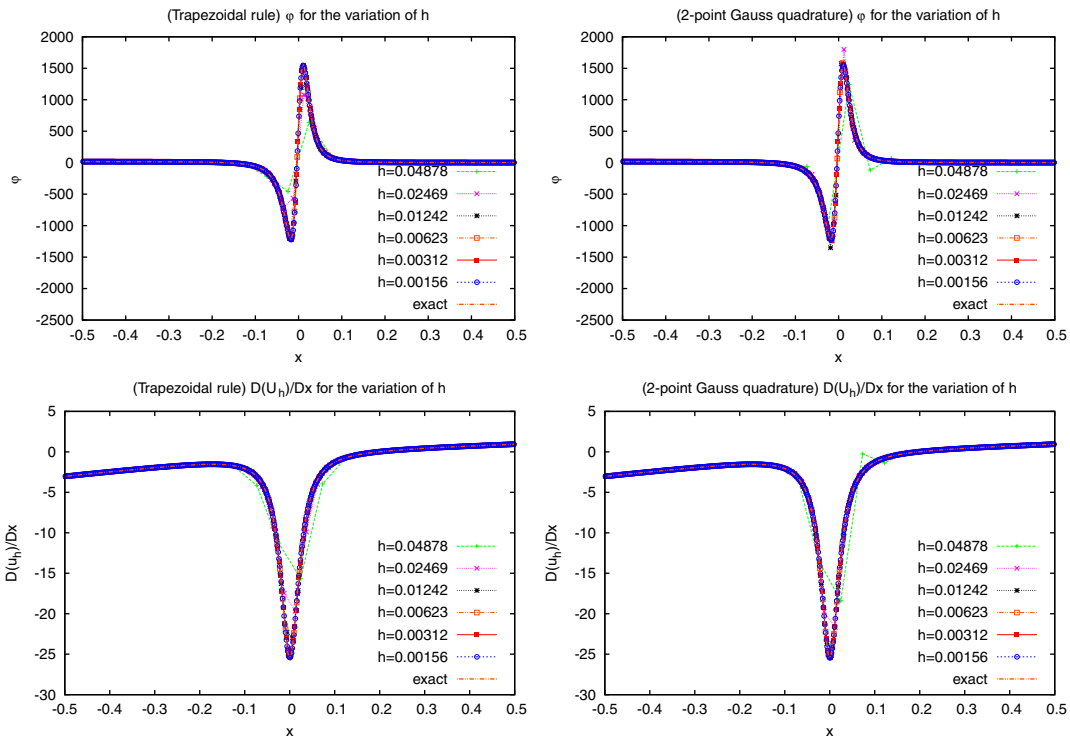

Figure 3. Results of the 1D example. Trapezoidal rule (left) and two-point Gaussian quadrature rule (right) of ϕ (top) and $\mathrm{d}u_h/\mathrm{d}x$ (bottom).

Boundary conditions are prescribed directly from the analytical solution as

$$u(x, y) = (1 + x^2 - y^2)\kappa(x, y) \tag{80}$$

The exact solution $u(x, y)$, as well as the coefficient $\kappa(x, y)$, experiences the steep gradient near the curve $(x+1)^2 + (y-1/2)^2 = \frac{5}{4}$. This problem set will be solved for the test problems of a circular domain (Section 6.1) and complex domains (Section 6.3). In addition, particularly for the test problem of non-smooth domain, we will solve the Laplace equation in Section 6.2 where the convergence behaviors are investigated for the case of singular solution.

Similar to the discretization in the previous 1D example, the axial lines are first constructed based on a large domain that completely covers the domain of solution. For instance, we take a large square domain enough to contain the whole domain Ω. By cutting off the axial lines at the boundary of Ω, we identify the axial lines for computation, which belong to the problem domain Ω.

For the convergence tests of 2D examples, the relative error of numerical solution u_h is calculated by the following measure as

$$\text{relative sup-error of } u_h = \frac{\sup_{(x_i, y_j) \in \Lambda^{\mathrm{o}}} |u(x_i, y_j) - u_h(x_i, y_j)|}{\sup_{(x_i, y_j) \in \Lambda^{\mathrm{o}}} |u(x_i, y_j)|} \tag{81}$$

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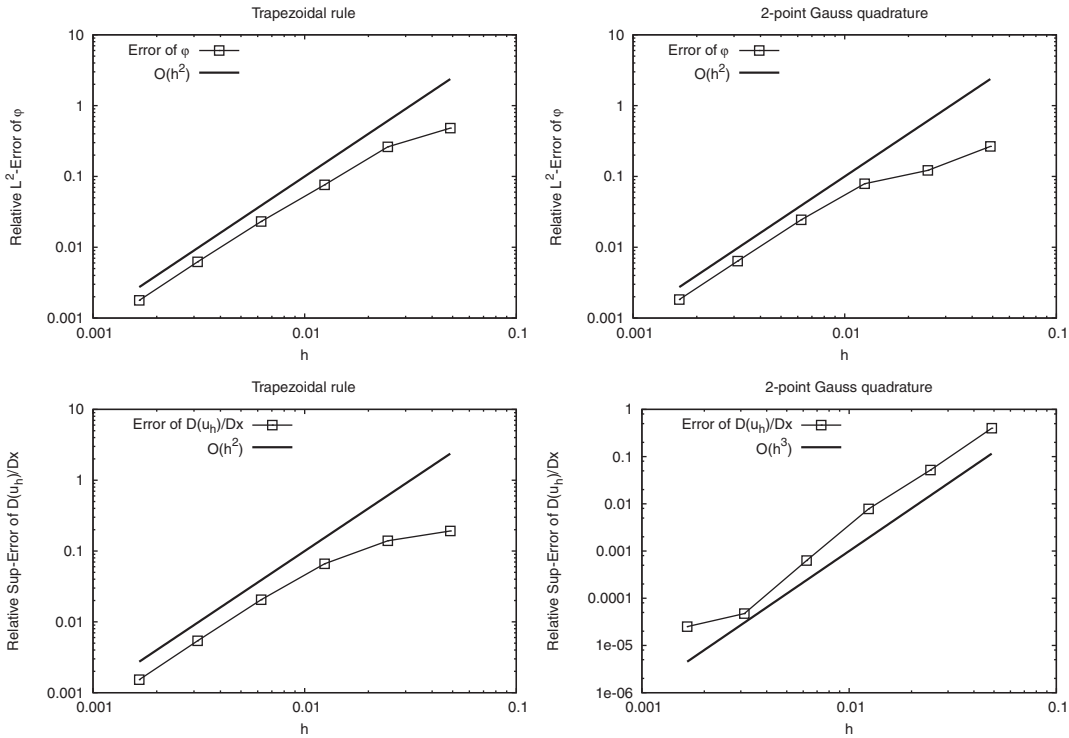

Figure 4. Results of the 1D example. Trapezoidal rule (left) and two-point Gaussian quadrature rule (right) of ϕ (top) and $\mathrm{d}u_h/\mathrm{d}x$ (bottom).

where u is the exact solution. For the derivatives, we measure the relative sup-error in the same manner.

6.1. Convergence test for a circular domain as an example of smooth domain

Let Ω be a circular domain with radius 0.5 and centered at the origin $(0, 0)$. The governing equation is given by Equation (78), and the boundary condition is induced from the exact solution, Equation (80). The axial lines, X_j's and Y_i's used in this example, are uniformly distributed inside the domain Ω, as shown in Figure 5. From the system of equations for ϕ addressed in Equation (70), we first solve for ϕ, and then obtain the numerical solutions, u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$ by the representation formula, Equations (43) and (44), and the axial line derivative, Equation (12). Figure 6 illustrates the numerical solutions in the circular problem domain Ω. The results are overlapped with the exact solution calculated in the larger square domain, for the purpose of comparison. We used five different discretized models to study the convergence. The numbers of x-axial lines are 20, 30, 40, 50, and 60. The numbers of y-axial lines are 20, 30, 40, 50, and 60. The numbers of base points are 332, 732, 1280, 1992, and 2876, respectively, for the five discretizations. In this paper, the error decay rate will be given in terms of the representative line space h for the evenly spaced x- and y-axial lines. The parameter h plays an equivalent role as the number of base points (N) through the relationship $h \approx (|\Omega|/N)^{1/d}$ where d is the space dimension and $|\Omega|$ stands for the volume of

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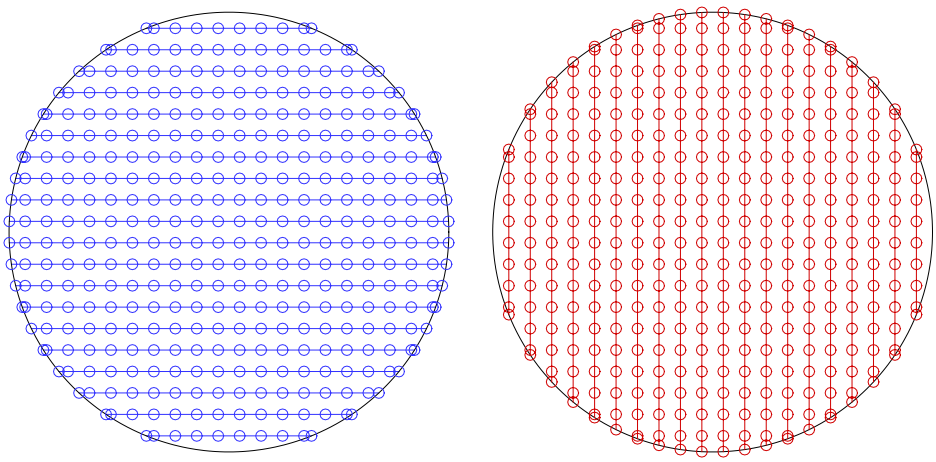

Figure 5. Axial lines, X_j's (left) and Y_i's (right), in a circular domain. The numbers of x- and y-axial lines and base points are, respectively, 20, 20, and 332 for this discretization model.

the problem domain Ω. The convergence results are given in Figure 7. The numerical solution $u_h(x, y)$ has the convergence rate of $O(h^2)$ for both numerical integration schemes. Interestingly, the second-order convergence is obtained also for the derivative of $u_h(x, y)$, even though the nodes are not regularly distributed in the vicinity of the boundary and the approximation for ϕ is even linear. This feature is not generally found in the finite difference method.

6.2. *Convergence test for the Lipschitz domain as an example of non-smooth domain*

We investigate the effect of the non-smoothness in the domain on the convergence of the numerical solution of the axial Green's function method, for which here we solve the Laplace equation. As a standard non-smooth domain, the well-known 2D Lipschitz domain is considered, which has a corner point on its boundary. Figure 8 illustrates the Lipschitz domain Ω having the corner point at the origin. The interior angle of the corner point is denoted by π/α, and we have $\frac{1}{2}<\alpha<\infty$. It is known that the solution $u(x, y)$ of the elliptic problem for this type of domain has a corner singularity. For example, if we impose the zero boundary value on two straight lines ($\theta=0, \pi/\alpha$), then a non-zero solution of the Laplace equation with such boundary condition asymptotically has the following singularity near the origin

$$u(x, y) \sim r^{\alpha} \sin(\alpha\theta) \quad \text{as } r \to 0 \tag{82}$$

where $r=\sqrt{x^2+y^2}$ and θ is the angle measured from the positive x-axis to the point (x, y) in the counterclockwise direction as shown in Figure 8. As a result, this singular behavior influences the regularity of the solutions. The leading singularity will thus be employed in the exact solution as

$$u(x, y)=r^{\alpha} \sin(\alpha\theta) \tag{83}$$

which satisfies the Laplace equation and the zero boundary values on $\theta=0$ and $\theta=\pi/\alpha$. Along the circular boundary of the domain Ω, we impose the value of the exact solution u for the boundary condition.

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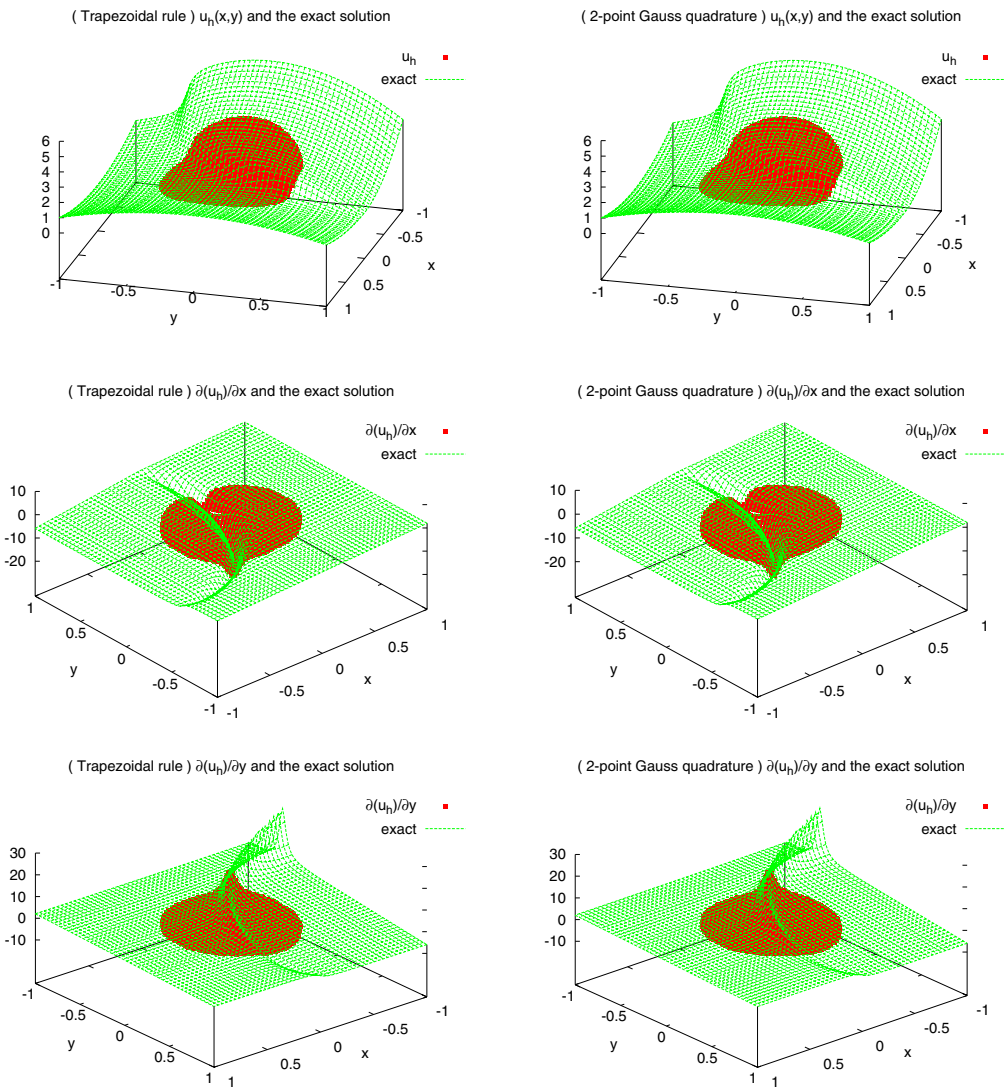

Figure 6. Numerical solutions of the circular domain problem and its exact solution. Trapezoidal rule (left) and two-point Gaussian quadrature rule (right) of u_h (top), $\partial u_h/\partial x$ (middle), and $\partial u_h/\partial y$ (bottom). The numbers of x- and y-axial lines and base points are, respectively, 60, 60, and 2876.

The regularity of the solution given in Equation (83) depends on the value α. If $\alpha \geqslant 2$ (i.e. the Lipschitz domain belongs to the 1st quadrant in Figure 8), then the solution $u(x, y)$ has at least continuous second-order derivatives up to the boundary. If $1<\alpha<2$ (the 2nd quadrant), the partial derivatives, $\partial u/\partial x$ and $\partial u/\partial y$, are Hölder continuous with exponent $\alpha-1$ up to the boundary, i.e. $u \in C^{1,\alpha-1}(\overline{\Omega})$, symbolically. The Hölder space $C^{k,\beta}(\overline{\Omega})$ for a non-negative integer k and an exponent β $(0<\beta\leqslant 1)$ consists of the functions of which kth-order partial derivatives are uniformly Hölder continuous with the exponent β in Ω (see [33]). In contrast, if $\frac{1}{2}<\alpha<1$ (the

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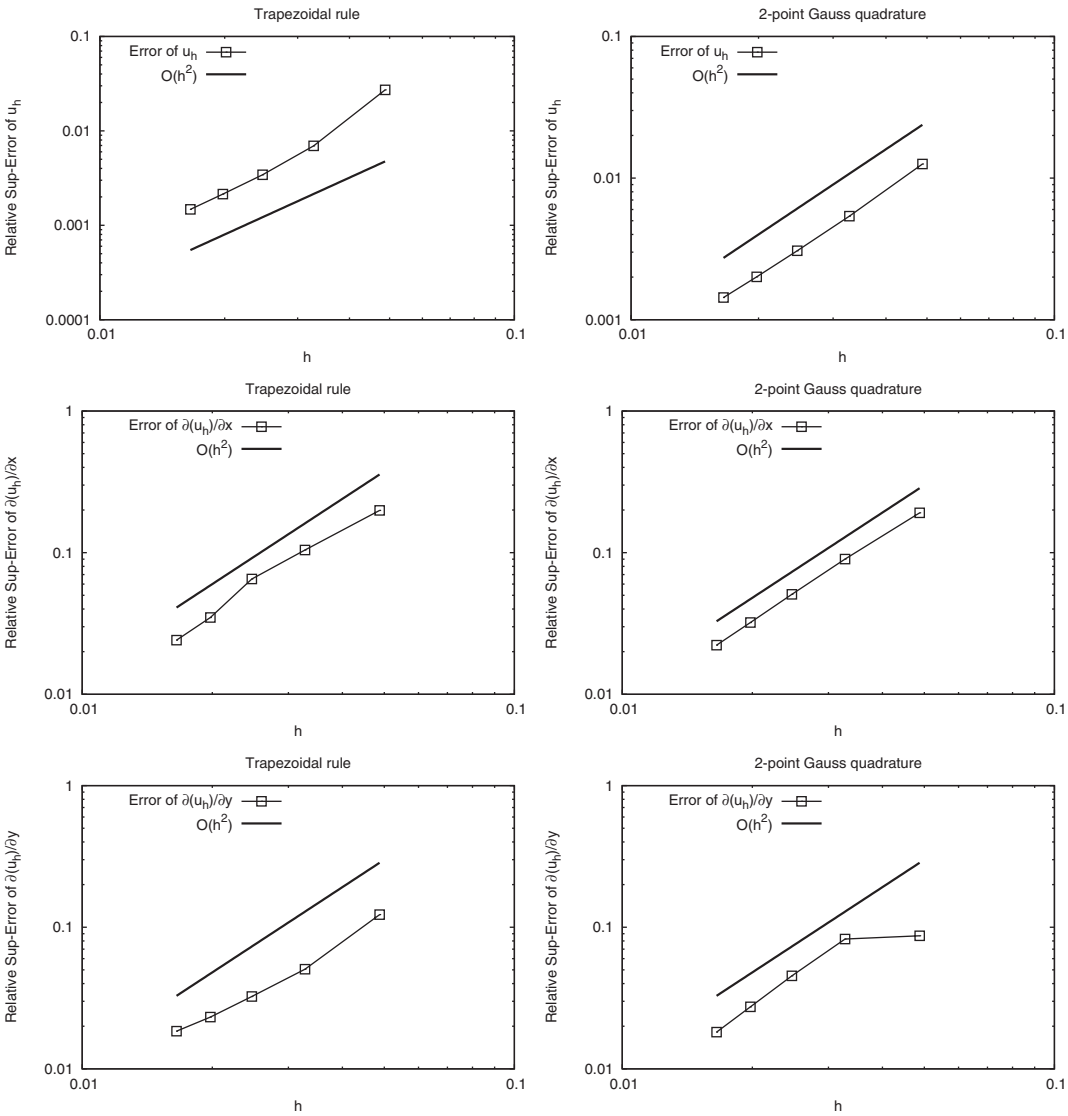

Figure 7. Convergence results of the circular domain problem. Trapezoidal rule (left) and two-point Gaussian quadrature rule (right) of u_h (top), $\partial u_h/\partial x$ (middle), and $\partial u_h/\partial y$ (bottom).

3rd or 4th quadrant), then we have $u \in C^{0,\alpha}(\overline{\Omega})$, which means that its derivative is not continuous up to the boundary. The limit case, $\alpha = 1$, corresponds to smooth boundaries, and it is thus excluded.

The convergence rates for the numerical solution u_h and $\partial u_h/\partial x$ of the axial Green's function method are illustrated in Figures 9 and 10, respectively. The figures show how the value α of the corner angle affects the convergence rate of the current method. The value α determines the order

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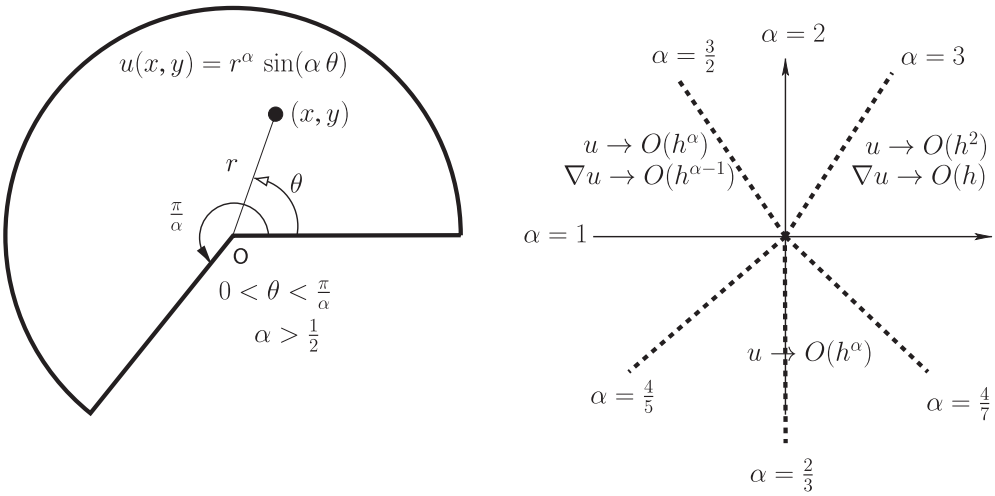

Singular solution on Lipschitz domains

Sup-error convergence rate for singular solutions

Figure 8. The Lipschitz domain for convergence test (left). The correlation of regularity (α) and convergence orders (right).

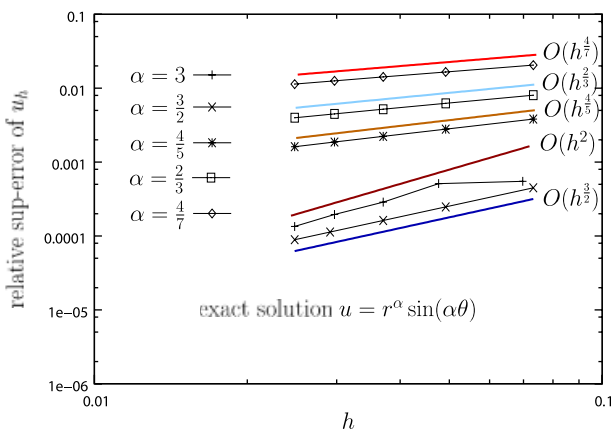

Figure 9. The dependence of convergence order (u_h) upon the solution regularities (α) in the Lipschitz domains.

of convergence for u_h and it is anticipated in the following manner:

$$\|u - u_h\|_{L^\infty} = O(h^{\min\{\alpha,2\}}) \tag{84}$$

However, contrary to the smooth boundary case, the order of convergence for derivatives of the numerical solution is reduced by one order, which is well depicted in Figure 10. The sup-error convergence for derivative is $O(h)$ when $\alpha \geqslant 2$ and $O(h^{\alpha-1})$ when $1<\alpha<2$.

In Figure 11, we illustrate the results of u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$ for the two representative cases of $\alpha = \frac{3}{2}$ and $\frac{4}{7}$ that, respectively, correspond to interior angles $2\pi/3$ and $7\pi/4$ at the origin.

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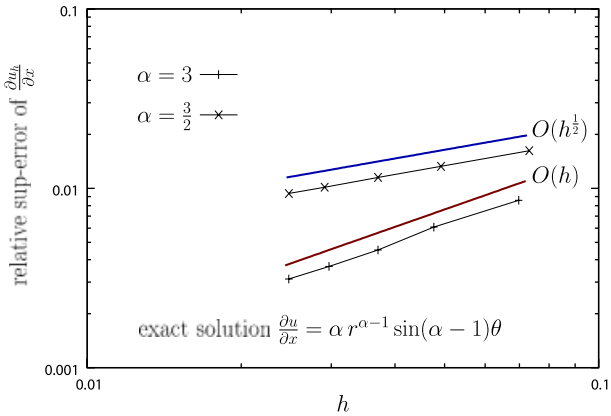

Figure 10. The dependence of convergence order ($\partial u_h/\partial x$) upon the solution regularities (α) in the Lipschitz domains.

From this numerical example of non-smooth domain, the proposed axial Green's function method is able to catch up the singularity effect, and therefore it properly represents the geometric property of the solution.

6.3. Complex domains

We now consider complex domains, such as a star-shaped domain and a spiral domain, in order to demonstrate that the current method can readily be applied to any type of geometry, which is one of the highlighted advantages of the method. As in the previous circular domain case, the equation to solve is Equation (78) with the boundary condition prescribed from the exact solution (80).

First, the star-shaped domain is considered. The number of x- and y-axial lines are 85 and 104, respectively, and the total number of base points (i.e. cross points) is 3042. The axial lines are regularly distributed by equal distance between them, being similar to the case of the circular domain. Figures 12 and 13 show the results of ϕ_h, u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$, together with their exact solutions. In the figures, the left and right columns are the results obtained by the trapezoidal rule and two-point Gaussian quadrature, respectively. The relative errors of ϕ_h, u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$ using the trapezoidal rule are 2.4×10^{-2}, 1.6×10^{-3}, 3.5×10^{-2}, and 2.3×10^{-2}, respectively, whereas those by two-point Gaussian quadrature are calculated to be 4.3×10^{-2}, 1.6×10^{-3}, 3.3×10^{-2}, and 2.2×10^{-2}.

For the same star-shaped domain, we also solve the problem with the randomly distributed axial lines. The random number generator produces x- and y-axial lines in each direction. We then cut them off to make the star-shaped domain and add the boundary points to each axial lines cut off. The resultant axial lines are shown in Figure 14. The numbers of x- and y-axial lines become 89 and 95, respectively, and we have total 3602 base points. The two-point Gaussian quadrature rule is employed as an integration rule. From the top in Figure 15, the numerical solutions, u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$, are compared with the corresponding exact solutions. The relative errors of u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$ are 1.0×10^{-2}, 1.4×10^{-1}, and 1.1×10^{-1}, respectively. We have measured computing time with different refinements for this example of star-shaped domain. In all cases, the preprocessing of axial-line discretization and the computation of conjugate gradient

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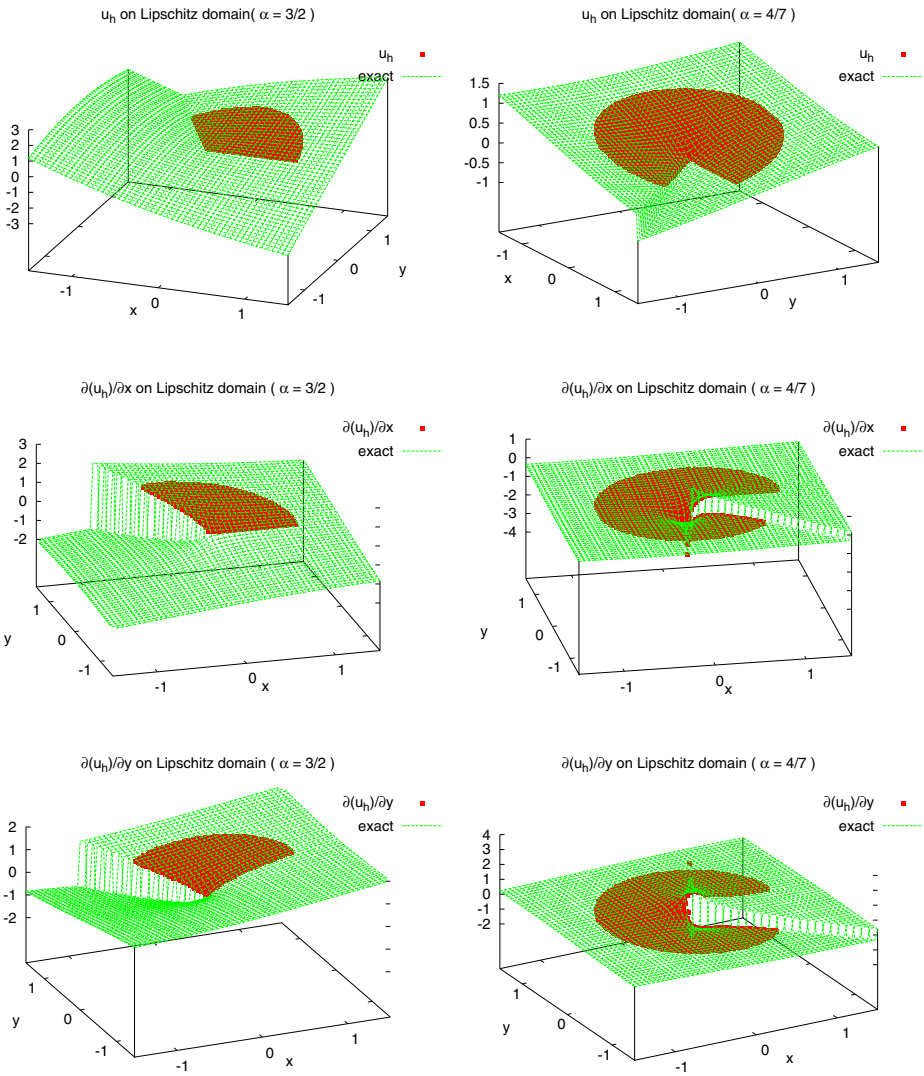

Figure 11. Numerical solutions and derivatives for $\alpha = \frac{3}{2}$ and $\frac{4}{7}$.

solver take a relatively small portion of the whole computation. Most of computing time (70–80%) is spent for building the discretized matrix system by numerical integrations.

Next, we consider a spiral-shaped domain for the same problem, Equation (78) with Equation (80), as shown at the top-left in Figure 16 where we plot the x- and y-axial lines, base points, and boundary point, all together to better illustrate the idea of axial discretization. The axial lines are constructed to be equispaced in each direction. In general, it is not easy for the finite difference method to achieve the second-order convergence in such a complex domain, unless the qualitative orthogonal grid is carefully constructed. On the other hand, in the axial Green's function method, the generation of axial lines to maintain the second-order convergence is much simpler than the

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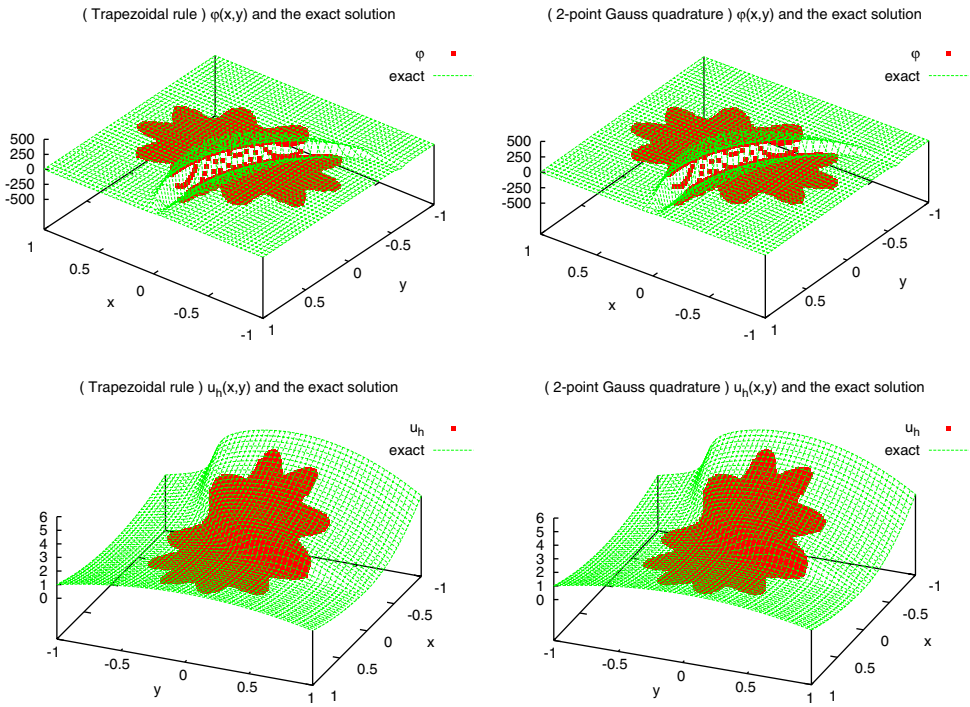

Figure 12. Results of the example of star-shaped domain. Trapezoidal rule (left) and two-point Gaussian quadrature rule (right) of ϕ (top) and u_h (bottom).

finite difference method, as demonstrated in this example of spiral domain. The numbers of x- and y-axial lines, and base points used for the discretization are, respectively, 155, 146, and 1634. Two-point Gaussian quadrature rule is employed. The results are given in Figure 16. The relative errors obtained are 1.25×10^{-3}, 2.05×10^{-2}, and 2.65×10^{-2}, respectively, for u_h, $\partial u_h/\partial x$, and $\partial u_h/\partial y$.

7. 3D NUMERICAL EXAMPLE

We finally examine the convergence rate of a 3D problem with a spherical domain $\Omega\subset\mathbb{R}^3$ that has a radius of $r=3$ and centered at the origin $(0,0,0)$. The elliptic partial differential equation to solve is

$$\nabla\cdot(\kappa(x,y,z)\nabla u(x,y,z))=f(x,y,z),\qquad (x,y,z)\in\Omega \tag{85}$$

where

$$\kappa(x,y,z)=1+x^2+y^2+z^2 \tag{86}$$

The boundary conditions are given directly from the analytical solution as

$$u(x,y)=\sin(x\,y\,z) \tag{87}$$

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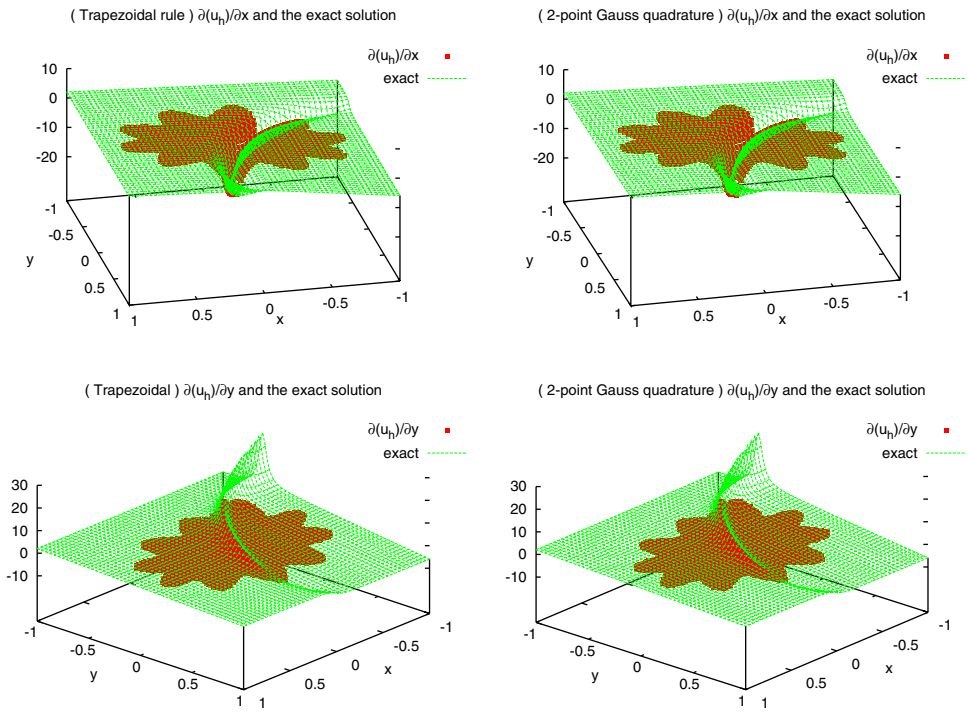

Figure 13. Results of the example of star-shaped domain. Trapezoidal rule (left) and two-point Gaussian quadrature rule (right) of $\partial u_h/\partial x$ (top) and $\partial u_h/\partial y$ (bottom).

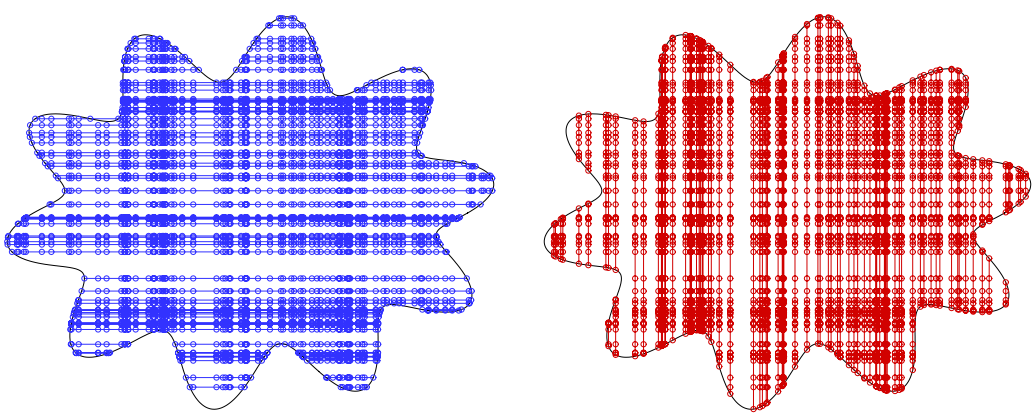

Figure 14. Irregularly distributed axial lines, X_j's (left) and Y_i's (right), in a star-shaped domain.

Figure 17 shows the boundary points and the base points of the spherical domain. The numbers of x-, y-, and z-axial lines are 1092, respectively, and the total base points are 27 016. Two-point Gaussian quadrature rule is employed. As shown in Figure 18, the second-order convergence rate,

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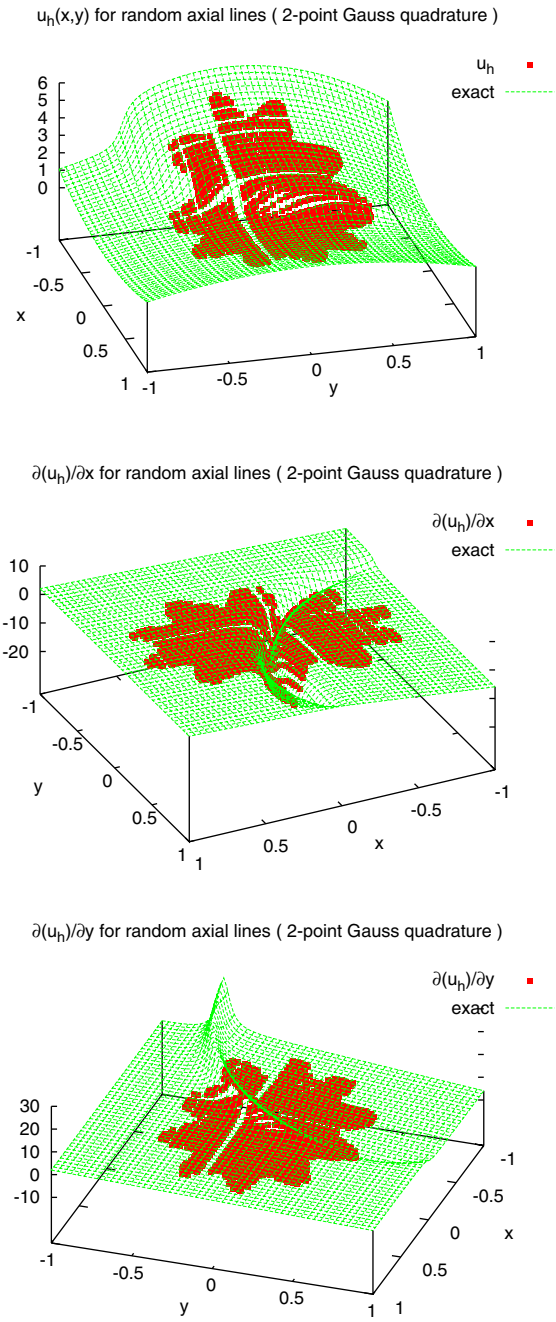

Figure 15. Results of u_h (top), $\partial u_h/\partial x$ (middle), and $\partial u_h/\partial y$ (bottom) for irregularly distributed axial lines.

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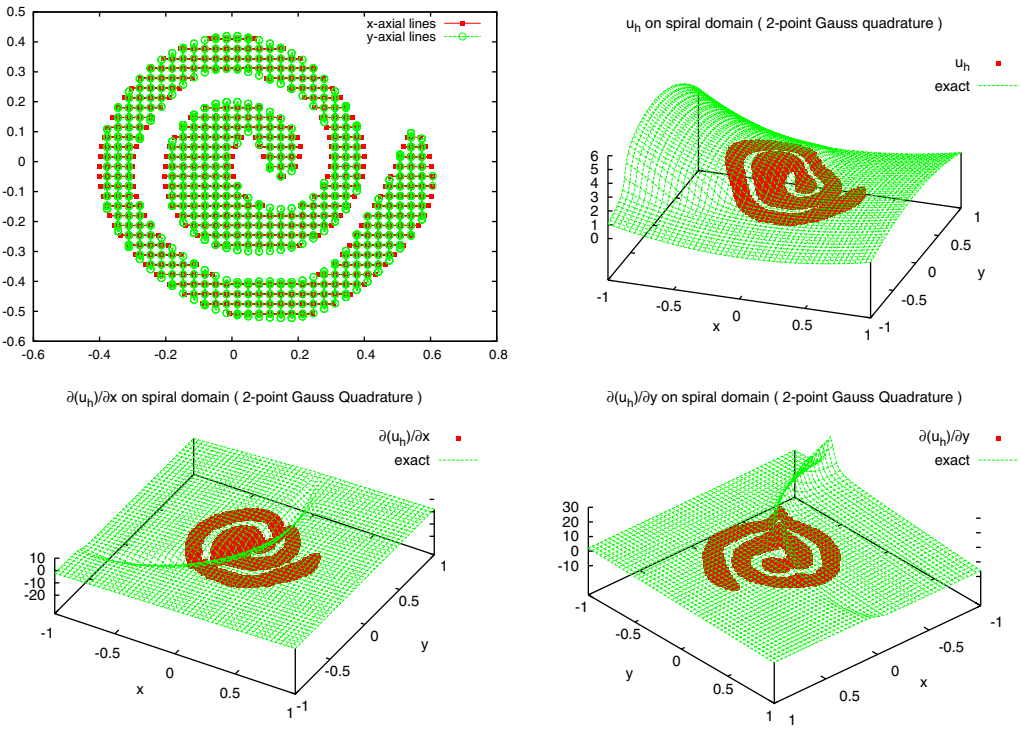

Figure 16. Axial lines, X_j's and Y_i's, and base points in a spiral domain (top-left), u_h (top-right), $\partial u_h/\partial x$ (bottom-left), and $\partial u_h/\partial y$ (bottom-right).

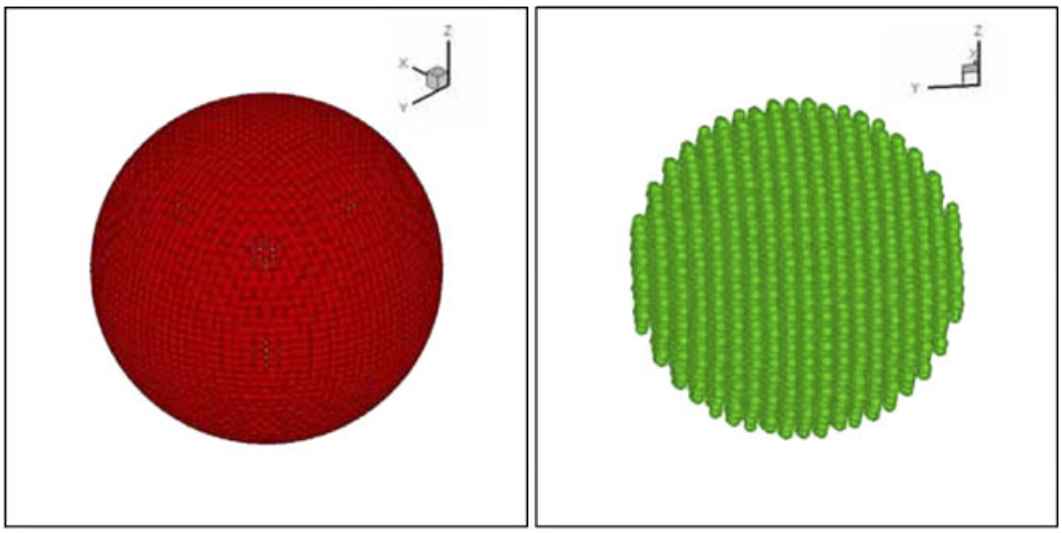

Figure 17. Boundary points and base points in a spherical domain of the 3D example.

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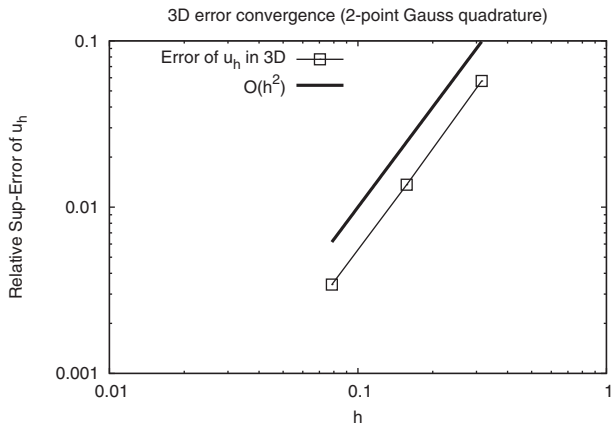

Figure 18. Convergence rate of the 3D example.

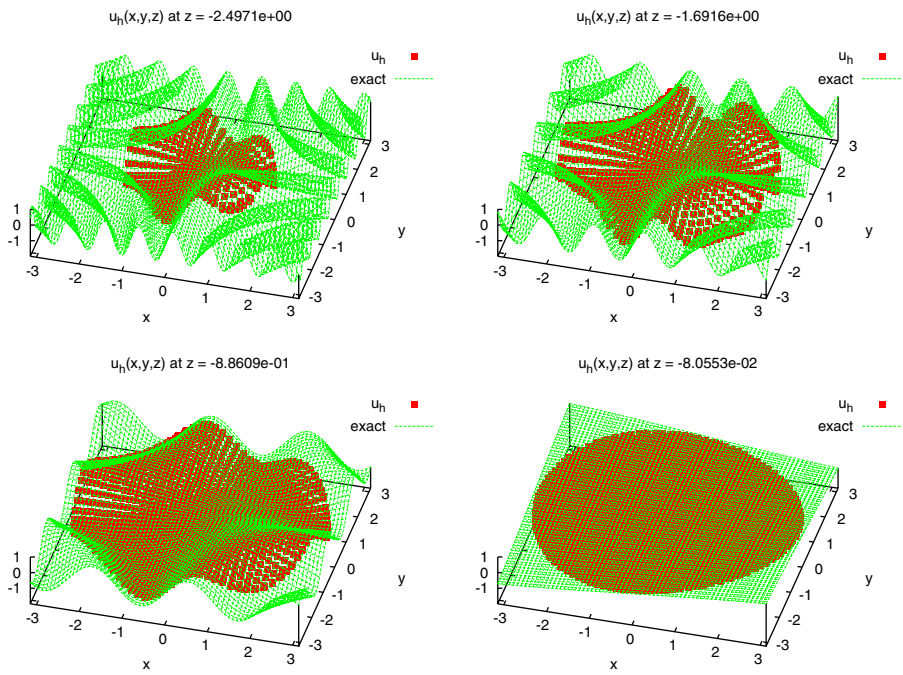

Figure 19. Sectional views of the numerical solution $u_h(x, y, z)$ in the 3D problem.

$O(h^2)$, still applies to the numerical solution, u_h, of this 3D example. The numerical solution $u_h(x, y, z)$ for $h = 2\pi/39$ is compared with the exact solution $u(x, y, z)$ in Figure 19. For clarity, sectional views of the solution along the z-direction are illustrated. The samples are taken at $z_1 = -2.4971$, $z_2 = z_1 + 5h$, $z_3 = z_2 + 5h$, and $z_4 = z_3 + 5h$. The error profiles, $|u - u_h|$, on these sectional planes are depicted in Figure 20.

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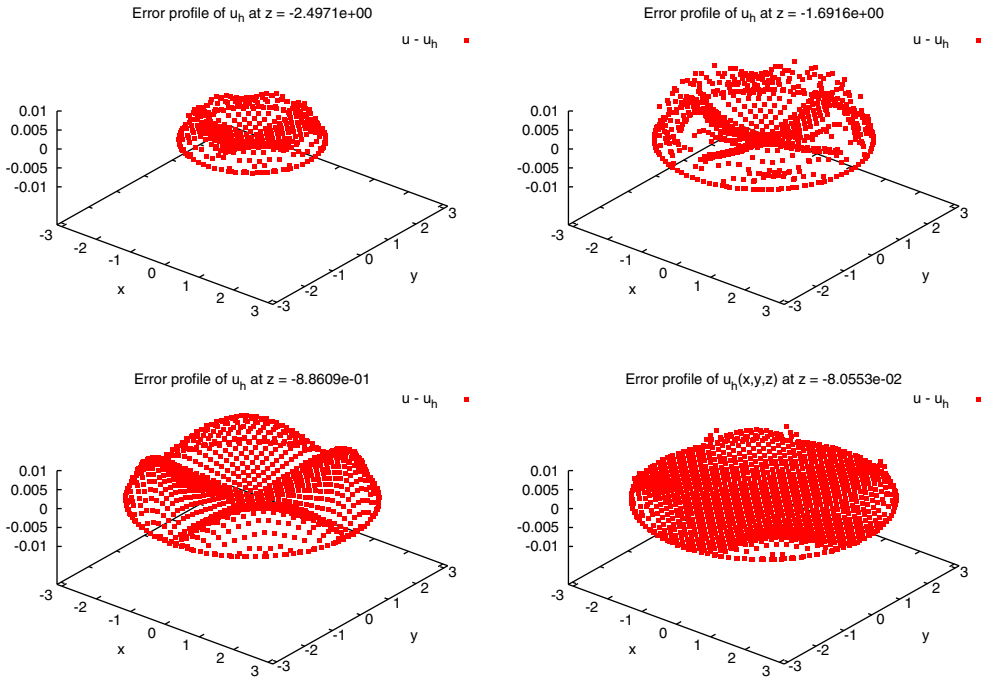

Figure 20. Sectional views of the absolute error of $u_h(x, y, z)$ in the 3D problem.

8. CONCLUDING REMARKS

We have presented a novel numerical method that can solve general multi-dimensional second-order elliptic partial differential equations by employing 1D Green's function method and simple axial discretizations. The multi-dimensional elliptic operators are decomposed into 1D counterparts in accordance with axial discretization of a complex solution domain. The axial Green's function method is then applied to the solution procedure, and the system of equations is obtained. The solutions of the system of equations are finally converted into the numerical solution of the original multi-dimensional elliptic boundary value problem. Through several numerical examples, we have shown that the method achieves the second-order convergence even for the cases of irregular axial discretization. The convergence behavior of the method obeys the standard rule in that the derivative is generally of one-order lower convergence than the solution. However, for the cases where the domain boundary is smooth and so is the boundary value, we observe much higher convergence results for the first derivative than usual in other numerical methods. The proposed method also contains the characteristics of integral schemes, which distinguishes it from the conventional finite difference method. We currently develop the method further for more complex problems that involve the discontinuity in solution. Future study includes various types of boundary value problems including Neumann boundary conditions. The potential flow problem, the Navier–Stokes equation, elasticity problems, hyperbolic differential equations, etc. are among examples in which the method can effectively be applied.

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